Practical 10: Contextual and Popup Menus

Source: MODULE 2.pdf | Source pages: 15–20

Aim

To develop and test an Android application demonstrating contextual and popup menus.

Steps to Perform the Practical

1. Create a project named MenuDemo.

2. Add the TextView and Popup Menu button in activity_main.xml.

3. Create the res/menu resource directory and add menu_options.xml with Edit, Delete and Share.

4. In MainActivity.java, register the TextView for a context menu and create a PopupMenu for the button.

5. Run the app. Long-press the TextView to open the contextual menu.

6. Tap Show Popup Menu and select each item. Confirm a Toast displays the selected item.

Result

The application is run and the expected behavior is verified according to the practical instructions and code.

Note: Use the package name and resource names from the source code consistently. If Android Studio reports a missing resource or class, check spelling and ensure all files are in the correct folders.

3. Develop an Android application implementing contextual menus and popup menus.

ProjectName: MenuDemo

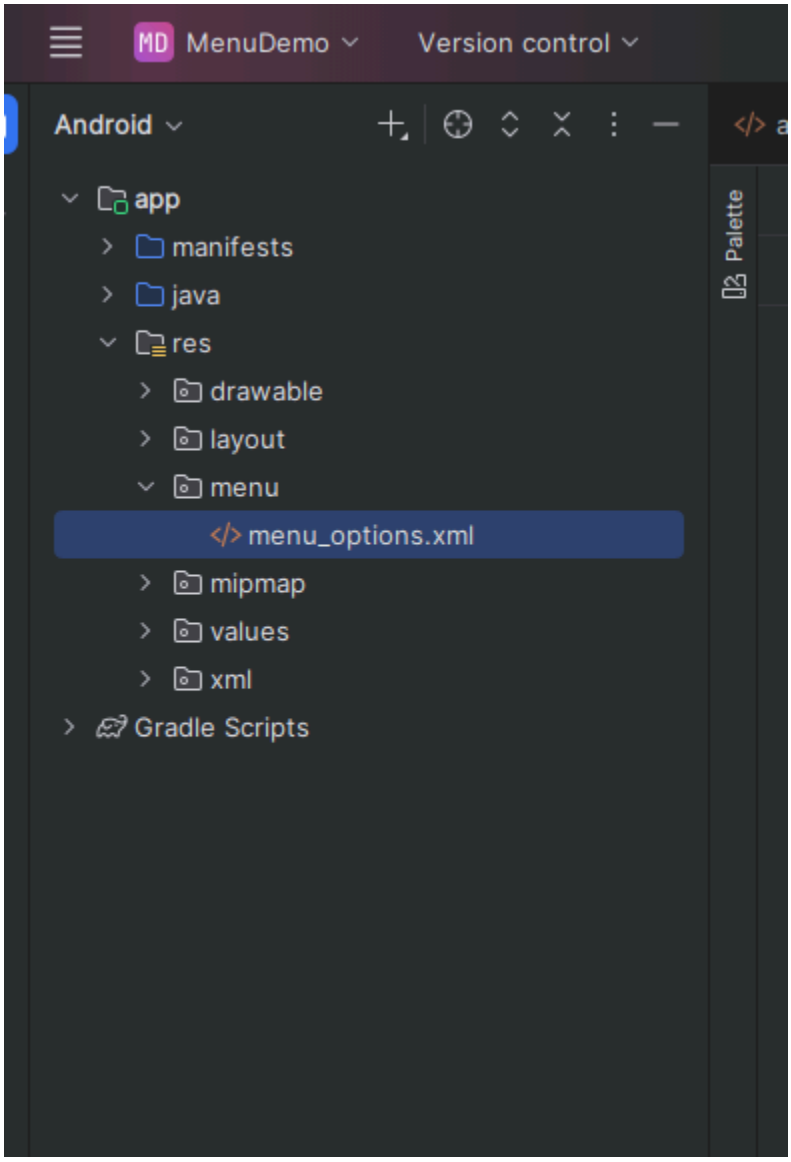
File Name: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:padding="20dp">
   <TextView
       android:id="@+id/txtContext"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="Long Press Me"
       android:textSize="24sp"
       android:padding="20dp"/>
   <Button
       android:id="@+id/btnPopup"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="Show Popup Menu"
       android:layout_marginTop="30dp"/>
</LinearLayout>
```

Create a Menu Resource File.
Right Click: res→ New → Android Resource Directory
Directory Name: menu
Resource Type: menu
Click OK
Right Click: menu → New → Menu Resource File

File Name : menu_options



File Name : menu_options.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<menu xmlns:android="http://schemas.android.com/apk/res/android">

   <item

       android:id="@+id/menu_edit" android:title="Edit"/>

   <item

       android:id="@+id/menu_delete"

       android:title="Delete"/>

   <item

       android:id="@+id/menu_share"

       android:title="Share"/>

</menu>
```

File Name: MainActivity.java

```
package com.example.menudemo;

import android.os.Bundle;

import android.view.ContextMenu;

import android.view.MenuItem;

import android.view.View;

import android.widget.Button;

import android.widget.PopupMenu;

import android.widget.TextView;

import android.widget.Toast;

import androidx.annotation.NonNull;

import androidx.appcompat.app.AppCompatActivity;
```

```java
public class MainActivity extends AppCompatActivity {


   TextView txtContext;

   Button btnPopup;


   @Override

   protected void onCreate(Bundle savedInstanceState) {

       super.onCreate(savedInstanceState);

       setContentView(R.layout.activity_main);


       txtContext = findViewById(R.id.txtContext);

       btnPopup = findViewById(R.id.btnPopup);


       // Register Context Menu

       registerForContextMenu(txtContext);


       // Popup Menu

       btnPopup.setOnClickListener(v -> {


           PopupMenu popupMenu = new PopupMenu(MainActivity.this, btnPopup);


           popupMenu.getMenuInflater().inflate(

                   R.menu.menu_options,

                   popupMenu.getMenu());
```

```
        popupMenu.setOnMenuItemClickListener(item -> {
            Toast.makeText(MainActivity.this,
                    item.getTitle(),
                    Toast.LENGTH_SHORT).show();
            return true;
        });

        popupMenu.show();
    });
}

// Context Menu Creation
@Override
public void onCreateContextMenu(ContextMenu menu,
                                View v,
                                ContextMenu.ContextMenuInfo menuInfo) {

    super.onCreateContextMenu(menu, v, menuInfo);

    menu.setHeaderTitle("Select Option");
    getMenuInflater().inflate(R.menu.menu_options, menu);
}

// Context Menu Click Handling
@Override
public boolean onContextItemSelected(@NonNull MenuItem item) {
```

```
        Toast.makeText(this,
                item.getTitle(),
                Toast.LENGTH_SHORT).show();


        return true;
    }
}
```

